

Pre-Analysis Plan

Does Satu Sehat API integration increase TB case notification by private primary care clinics in Bandung, Indonesia? A proof-of-concept cluster-randomized trial

Draft Pre-Analysis Plan. Stage 1 of a two-stage research program.

Abstract

Indonesia carries the world's second-largest tuberculosis burden, and its private primary care sector — where most patients first seek care — accounts for a small fraction of cases reported to the national surveillance system (SITB). This study tests whether providing private clinics with automated reporting infrastructure, rather than behavioral training, can shift notification rates. The intervention connects a clinic's existing electronic medical record to the national Satu Sehat platform via FHIR R4 API, so that entering a TB ICD-10 code automatically pushes a notification to SITB. The design is a cluster-randomized trial with the clinic as the unit of randomization, stratified by subdistrict, clinic size, and prior notification history. The primary outcome is total SITB notifications per clinic over twelve months. A pre-specified mechanism check examines the ratio of TB-specific to broad respiratory ICD-10 codes to distinguish friction reduction from coding avoidance. The study is framed as proof-of-concept (Stage 1): the eligible sample is constrained by current EMR adoption among Bandung private clinics, and statistical power for a 20-percentage-point effect ranges from approximately 52% to 62% under realistic enrollment scenarios. The study is therefore not powered for confirmatory inference. Its purpose is to establish whether the technical integration is achievable at clinic level, to detect a signal if a large effect exists, and to identify whether administrative friction or active avoidance binds. Effect size estimates and operational findings will inform the design of a fully powered confirmatory trial.

1. Introduction

1.1 Background: the reporting process and its costs

A private general practitioner in Bandung who diagnoses tuberculosis is required to report the case to SITB, the national TB information system. The provider must first record the diagnosis in the clinic's own system, an EMR or paper register, as part of routine clinical care. To complete the notification, the provider must then open a separate platform: the SITB web portal or, for individual case-based notifications by GPs and private clinics, the WiFi TB mobile application [3]. They must locate or create a record for the patient in that system, manually re-enter the diagnostic information already entered in the clinic's own record, and submit. The mobile-application and web portal pathways both require a working internet connection and authenticated login.

Three features of this process matter for the analysis. First, the time cost is unpaid: under Indonesia's national health insurance scheme (JKN), private primary care providers receive payment via capitation, with no additional payment tied to TB diagnosis or notification [4]. There is no private financial benefit to reporting. Second, the social benefits — surveillance, contact tracing, drug-resistance monitoring, resource allocation — are of higher interest to the public health system and the community than the clinic that bears the cost. This is the structure of a positive externality combined with administrative friction, and it is the part of the underreporting problem this study addresses.

Qualitative work on private-provider reporting in Indonesia has documented a structure of barriers consistent with this characterization. In Yogyakarta, Kurniawati and colleagues found that private practitioners did not notify TB cases primarily because of a lack of policy knowledge, that mandatory notification and its potential penalties were felt as burdensome, and that providers held ethical concerns about patient privacy [5]. Bandung and other Indonesian cities report additional friction: low awareness of national guidelines, inadequate diagnostic capacity at the point of first contact, and weak engagement with the National TB Programme [6,7]. Of these, the present study targets the procedural and unfamiliarity components. It does not address financial misalignment under JKN, confidentiality concerns, or diagnostic uncertainty about whether a case meets TB criteria.

1.2 Motivation

Indonesia is one of a small group of high-burden countries where TB incidence in 2024 was higher than in 2015, alongside Myanmar and the Philippines [1]. WHO estimates approximately 1.09 million TB cases in Indonesia annually, with roughly 125,000 deaths attributed to TB [1]. Underreporting is concentrated in the private sector. Lestari and colleagues' patient-pathway work in Bandung shows that approximately 75% of patients with eventual TB diagnoses first sought care at an informal or private provider [8]; nationally, Surya and colleagues reported that the private sector contributed only 9% of notifications in 2015 despite providing approximately 60% of outpatient care [9]. Aggregate underreporting has improved substantially: the 2016–17 National TB Inventory Study estimated 41% overall underreporting, while the 2023 repeat inventory study estimated overall underreporting at 15.6% (95% CI: 13.4–18.2%), a 2.6-fold reduction [2]. Within that aggregate, however, underreporting in private clinics remains 28% and in laboratories 55% [2]. Aggregate progress has been substantial. The private primary care sector remains the weakest link.

Underreporting matters for two reasons beyond the surveillance numbers themselves. First, transmission. A smear-positive TB patient can transmit infection to an estimated 10–15 close contacts annually if untreated, a figure widely cited in the case-finding literature [10]. The point is that the contact-tracing infrastructure cannot be activated for cases the system does not know exist. Second, cost. Allocation across districts and intervention modalities depends on surveillance data. Underreported districts are mis-allocated.

The study site is Bandung City, West Java. The Bandung Municipal Health Office's 2021 health profile identified 80 community health centres (Puskesmas), 358 private clinics, 22 secondary-level hospitals, and 15 tertiary-level hospitals operating in the city [11], a snapshot used in subsequent studies of the local TB system [12]. The 358 private clinics are the group of interest for the present study. The enumeration will be updated against current Bandung Dinas Kesehatan records before randomization.

1.3 Literature and existing programs

Indonesia has implemented several private sector engagement strategies for TB. The District-Based Public-Private Mix (DPPM) program operates nationally and coordinates between district health offices and participating private providers [13]. Clinics already enrolled in DPPM are excluded from this study to avoid contamination with an active engagement program.

The closest direct precedent is the INSTEP program in Bandung. The INSTEP1 feasibility study tested an intervention package of TB training, the WiFi TB mobile application, and follow-up visits with private practitioners across five Community Health Centre catchment areas [14]. Of 87 PPs registered in those catchments, 27 (31%) were interviewed at baseline, 17 attended training, and 12 had the WiFi TB application successfully installed; the remaining 5 had phones that did not support the application [14]. INSTEP2, registered as NCT04187313, is a cluster-randomized trial of a multi-component behavioral package in Bandung CHC catchments [15]. Clinics that participated in INSTEP2 are excluded from the present study. Sensitivity analysis will test the inclusion of those clinics, since exclusion may itself select on prior engagement.

On the technical side, two recent studies inform the design. Heryawan and colleagues conducted a content analysis of Satu Sehat developer-hub discussions and identified server issues (61 instances), data mapping issues (37), and FHIR profile management issues (9) as the principal pain points reported by integration teams [16]. Pratiwi and colleagues assessed SIMRS-SATUSEHAT integration readiness across 11 Yogyakarta hospitals and identified limited dedicated IT staff with interoperability skills, the absence of customized integration protocols, and the lack of earmarked budgets as recurring constraints in the integration process [17]. Satrio and Aqid examined EMR adoption in the Indonesian primary-care customer network of a major EMR vendor and report cumulative adoption equivalent to 8.9% of an estimated 39,852 eligible primary care facilities (3,533 facilities) over a 33-month observation window [18]. The 8.9% figure is the principal national benchmark used to bound the eligible sample for this study and is discussed in Section 2.2.

On expected effect sizes, Dewan and colleagues reviewed public-private mix projects in India and reported wide variation in case-notification effects across the projects with available data, providing a rough range of the kinds of effects multi-component PPM programs have produced in other LMIC settings [19]. Those estimates concern multi-component behavioral programs and

are not directly comparable to the infrastructure-only intervention tested here. They are used in Section 5 only as rough anchors.

2. Experimental Design

2.1 Intervention

Treatment clinics receive three components, all delivered once at the start of the trial period except the helpdesk, which is available throughout.

API integration setup. A research team health informatics specialist visits each treatment clinic and connects the clinic's existing EMR or clinic management system to the Satu Sehat FHIR R4 platform via API. Satu Sehat is built on HL7 FHIR standards [16] and is mandated as the national health-data integration backbone under Permenkes No. 24/2022 [20]. Once the connection is verified, entering an ICD-10 TB code (A15–A19) in the clinic system triggers an automated FHIR resource push to SITB; no additional step is required from the provider beyond ICD code entry that is already part of the clinic's billing or recordkeeping workflow.

One-time ICD-10 coding session. The same specialist conducts a brief session with clinic staff on correct entry of TB codes A15–A19. This is not a behavioral training program. It is technical guidance restricted to the single coding action that triggers the automated notification.

Technical helpdesk. A contact (phone and messaging) is available to treatment clinics throughout the twelve-month follow-up for technical issues: API failures, login problems, mapping errors. The helpdesk does not provide clinical advice or behavioral encouragement.

The intervention deliberately omits behavioral training on TB management, financial incentives, supervision visits, and any component targeting provider motivation or patient confidentiality. This is the proof-of-concept logic. If the friction component alone moves notifications, there is a policy implication. If it does not, the binding constraint is elsewhere.

Heryawan and colleagues' catalog of Satu Sehat developer pain points [16] implies that silent failures will occur. API call logging and a retry protocol are built into the integration to detect and record these events. Technical failure events are logged separately and are not counted as non-notifications in the main analysis. A pre-specified sensitivity analysis excludes clinic-months with documented technical failure.

2.2 Enrollment criteria and sampling strategy

The sampling frame is the 358 private clinics enumerated in the Bandung Municipal Health Office 2021 profile [11]. Before randomization, the research team will update the list against

current Dinas Kesehatan records to account for clinic openings, closures, and changes in operational status.

Clinics are eligible if they are private primary care facilities operating in Bandung City, have been in operation for at least twelve months prior to enrollment, use an EMR or clinic management system compatible with Satu Sehat, and are not currently enrolled in DPPM or another TB notification improvement program. Clinics are excluded if they are paper-based and unwilling to adopt a digital system, use an incompatible proprietary system and are unwilling to switch, are enrolled in DPPM, or participated in INSTEP2. Sensitivity analysis will test the inclusion of INSTEP2 clinics.

The eligibility criteria carry a substantial external validity cost. Restricting enrollment to EMR-compatible clinics selects a more digitally advanced subset of the private sector. The Satrio and Aqid analysis of one major EMR vendor's customer network estimates national adoption at 8.9% of eligible primary care facilities [18]; Bandung's adoption rate likely exceeds this given local infrastructure, but the precise figure requires Dinas Kesehatan records and must be obtained before sample size is finalized. Applied naively to the 358 mapped clinics, national-rate adoption implies roughly 30–50 immediately eligible clinics. Findings from this study will not generalize to paper-based clinics, which constitute a separate and larger policy challenge addressed by other instruments.

2.3 Assignment protocol

The unit of randomization is the clinic. Patient-level randomization is not feasible because a treated provider's reporting behavior changes for all patients seen, not only those nominally enrolled in the trial.

Allocation is by computer-generated random sequence, stratified on three variables. First, subdistrict: clinics in the same neighborhood share patient populations and TB exposure rates, and stratification on subdistrict prevents geographic confounding. Second, clinic size: larger clinics see more TB and an imbalance would systematically advantage one arm. Third, prior notification history, dichotomized as any SITB record in the twelve months preceding baseline versus none: prior reporters are more likely to report again, and balancing on this prevents the treatment arm from being stacked with already-compliant clinics. Equal allocation is used within strata.

Allocation is performed after the baseline survey is complete and the data are locked. Allocation concealment is maintained until that point. Because the intervention requires an in-person setup visit, providers cannot be blinded to assignment after intervention begins; outcome ascertainment uses administrative SITB records, which removes provider self-report from the primary outcome but does not eliminate the possibility of differential behavior conditional on assignment. Reporting of trial methods and results will follow the CONSORT extension for cluster randomized trials [21].

3. Data

3.1 Data sources

The primary data source is SITB administrative records, accessed through a formal data-sharing agreement with the Bandung City Dinas Kesehatan and the Ministry of Health. SITB is the same system used by the national TB program for routine surveillance; using it as the outcome measure means the study evaluates the intervention against the metric the policy itself targets. No additional outcome data collection from clinics is required for the primary outcome.

Three secondary data sources are used. First, a baseline survey of enrolled clinics, conducted before randomization, captures clinic size, EMR system in use, prior notification history, staff composition, and patient volume. The survey instrument is being developed and is not finalized. Second, ICD-10 coding records from clinic systems or claims data, used for the mechanism check described below. Third, technical logs from the API integration, used for the failure-event sensitivity analysis.

Two practical data limitations bear on the analysis. SITB completeness varies across clinic types, and notifications can be filed via multiple channels (web portal, mobile application, automated API push) that are not always uniformly indexed. The primary outcome counts notifications attributable to a clinic by clinic identifier in SITB; if multiple identifiers map to one clinic, deduplication will be done before analysis using the baseline clinic registry. ICD-10 coding access is conditional on each clinic's willingness to share records; coverage of the mechanism check will likely be incomplete.

3.2 Outcome construction

Primary outcome. Total TB notifications submitted to SITB by clinic i over the twelve-month follow-up period. The unit of analysis is the clinic; each clinic contributes one observation, which is the sum of notifications across the twelve follow-up months. This specification permits direct comparison to the approximately 4% control-arm benchmark for private GP/clinic SITB participation derived from the Stop TB Partnership's digital TB surveillance assessment for Indonesia [22] and avoids any intra-cluster correlation adjustment because there is no clustering below the clinic level under this aggregation.

Secondary outcome. Days from diagnosis to SITB notification (timeliness), computed for cases in which both the diagnosis date and the notification date are recorded in SITB. Because the API integration triggers on code entry, the expected effect on timeliness is mechanical: a treatment effect on this outcome of near zero days post-diagnosis is consistent with the

intended mechanism, and an effect that does not collapse the gap would suggest the integration is not behaving as designed.

Robustness specification. A monthly clinic-level panel with twelve observations per clinic. This captures the time pattern of any effect — whether notifications increase immediately after API setup, build over time, or display any other patterns. This is a secondary specification, not the primary analysis.

Mechanism check (pre-specified). The ratio of TB-specific ICD-10 codes (A15–A19) to broad respiratory codes (J00–J99) per clinic, compared across arms before and after the intervention, using clinic records or claims data where available. The interpretation is the central inferential payoff of the design:

- If treatment clinics show declining TB-specific coding after API installation, avoidance is active: providers are coding around the automatic trigger to avoid the notification.
- If TB-specific coding rates are stable or rising, the intended mechanism, friction reduction, is the dominant story, conditional on a positive effect on the primary outcome.

4. Empirical Strategy

Two specifications are pre-specified: a primary clinic-level analysis on the twelve-month total, and a secondary monthly panel that exploits within-clinic time variation. Both estimate the intent-to-treat (ITT) effect; the complier average causal effect (CACE / LATE) is reported as a secondary analysis using assignment as an instrument for verified API integration.

4.1 Primary specification

The primary specification is a clinic-level OLS regression of total notifications on treatment assignment and baseline covariates:

$$Y_i = \alpha + \beta \cdot T_i + \gamma' \cdot X_i + \varepsilon_i$$

where Y_i is the total number of SITB notifications submitted by clinic i over twelve months; T_i is an indicator equal to one if clinic i was assigned to API integration and zero otherwise; X_i is a vector of baseline covariates including clinic size, subdistrict fixed effects, and a binary indicator for prior notification history; and ε_i is the residual. The coefficient β is the ITT estimate of interest. Standard errors are heteroskedasticity-robust. No intra-cluster correlation adjustment is applied at this level: the outcome is constructed at the clinic level and there is no sub-clinic clustering by construction.

The covariates X_i correspond to the stratification variables. Subdistrict fixed effects also absorb any geographic confounders correlated with both treatment exposure and outcome.

4.2 Secondary specification

The secondary specification uses the monthly clinic-level panel:

$$E[Y_{it} | T_i, \alpha_i, \delta_t] = \exp(\beta \cdot T_i + \alpha_i + \delta_t)$$

estimated as a fixed-effects Poisson regression. Y_{it} is the count of notifications by clinic i in month t . α_i is a clinic fixed effect, absorbing all time-invariant clinic characteristics including baseline reporting propensity. δ_t is a month fixed effect, absorbing common time trends — seasonal variation in TB presentations, national-level program activity, and any system-wide changes to SITB during the trial. Standard errors are clustered at the clinic level. Poisson is preferred to a linear model here because the monthly count is a small integer with mass at zero and the conditional mean is constrained to be positive; the Poisson fixed-effects estimator is consistent for the conditional mean function regardless of the underlying distribution of Y_{it} and does not require the variance assumption to hold.

The secondary specification will be reported alongside the primary in the main results table. It is informative about the timing of the effect: an immediate post-setup increase that persists is consistent with mechanical friction reduction; a gradual ramp suggests a behavioral or learning component; a transient spike followed by reversion suggests novelty or a ceiling effect.

4.3 Intent-to-treat and complier average causal effect

The intent-to-treat analysis is primary. All enrolled clinics are analyzed in the arm to which they were randomly assigned, regardless of whether the API integration was successfully completed and verified. ITT preserves the experimental contrast and answers the policy-relevant question: what happens if a program offers integration to a population of eligible clinics?

As a secondary analysis, the complier average causal effect on the subset of treatment clinics for which the API connection was successfully verified is estimated by two-stage least squares using assignment as an instrument for verified integration. This isolates the effect of actual exposure to a working integration, conditional on the standard exclusion restriction (assignment affects notifications only through verified integration). The exclusion restriction is plausible here, but it is not airtight. Assignment includes the in-person setup visit and the coding session, which could in principle have a small behavioral effect even when the API connection itself fails. CACE estimates are therefore reported with this caveat and are not used as the primary policy estimand.

4.4 Multiple comparisons and pre-specification

The primary outcome is a single quantity: total notifications per clinic. The mechanism check produces a second test on the coding ratio. Timeliness is reported descriptively. No multiple-comparison adjustment is applied across these because they answer distinct questions; results will be reported separately. Robustness specifications (the monthly panel, the technical-failure sensitivity, the INSTEP2-inclusion sensitivity) are reported to assess fragility.

5. Power Calculations

The eligible sample is constrained by EMR adoption. The resulting power is below the conventional 80% threshold for the effect sizes considered policy-relevant.

The benchmark control rate is the share of private GP/clinic facilities in Indonesia that submitted any TB data to the national database in a given year. The Stop TB Partnership's Indonesia digital TB surveillance assessment reports that in 2020, 4% of GPs/private clinics across Indonesia reported TB data, compared to 61% of private hospitals [22]. Surya and colleagues' 2015 estimate — the private sector providing approximately 60% of outpatient care but contributing only 9% of notifications nationally [9] — is broadly consistent with the magnitude of this gap, and the 4% private GP/clinic figure is used as the control-arm benchmark for power purposes.

The effect size considered policy-relevant is a 20 percentage point increase in the share of clinics submitting at least one notification annually. Below this threshold, the integration cost would not plausibly be justified at scale relative to alternative uses of public health resources.

Power is computed for a two-sample test of proportions at $\alpha = 0.05$, two-sided, with equal allocation between arms:

Total clinics (N)	Approximate power
30	~52%
40	~62%
60	~74%
80	~82%

With the enrollment constraint of approximately 30–50 immediately eligible clinics, the achievable power for the primary scenario is in the range of 52% to 62%. This is insufficient for

confirmatory inference on the primary outcome and is acknowledged as such; running the study at this scale is justified only by its proof-of-concept purpose. The expected outputs from this design are: a point estimate and confidence interval for the effect size, which will inform the sample size calculation for a subsequent confirmatory trial; the mechanism check, which has its own interpretive value irrespective of power on the primary outcome; and operational findings about whether clinic-level FHIR integration is technically feasible at scale.

Power calculations will be re-verified with finalized stratification parameters before pre-registration. The dominant uncertainty is the local EMR adoption rate, which determines the eligible n . Obtaining a precise local figure from Bandung Dinas Kesehatan is a prerequisite for finalizing the sample size target.

6. Timeline and Costing

6.1 Timeline

The trial spans approximately seventeen months from preparatory work to write-up, with twelve months of post-randomization follow-up.

1. Months 0–2: update the 2021 clinic enumeration [11] against current Dinas Kesehatan records; conduct baseline survey across enrolled clinics; obtain SITB data-sharing agreement and ethics approvals.
2. Month 2: stratified randomization with allocation concealment maintained through this point.
3. Months 3–4: technical setup in treatment clinics. Per-clinic setup time is estimated to be on the order of two to four hours (a working assumption to be validated against actual setup logs from the first wave of treatment clinics, since no published study to our knowledge has measured per-clinic FHIR-to-SITB setup time at the primary care level). Under this assumption, approximately twenty person-days of technical labor are required across forty clinics.
4. Months 4–15: twelve-month follow-up. SITB data are pulled monthly. Helpdesk active throughout.
5. Months 15–17: data analysis and write-up.

6.2 Costing

Technical setup (dominant cost driver). The setup requires a health informatics specialist with demonstrated experience in FHIR R4 integration and the Satu Sehat platform [16]. Per-clinic work involves configuring API credentials, mapping clinic-specific diagnosis codes to ICD-10 A15–A19, writing or configuring an integration script, and verifying the connection in a sandbox environment before going live. The Pratiwi et al. assessment of SIMRS-SATUSEHAT readiness identifies limited dedicated IT staff with interoperability skills and the absence of customized integration protocols as recurring constraints in Indonesian integration projects [17], which suggests that the marginal cost of the specialist role will be substantial relative to other budget lines. The total cost depends on which staffing model is feasible. Under a university partnership — for instance with Universitas Padjadjaran, which has health informatics capacity in Bandung and ran INSTEP2 [15] — technical staff may be available at academic labor rates as part of a research collaboration. Under an external-consultant model, rates reflect Indonesian market rates for FHIR-capable health IT staff. These rates have not been formally surveyed for this PAP and require direct quotes from qualified implementers before budgeting. Resolving the staffing model is the first costing prerequisite.

Transport. Within-Bandung visit costs are modest relative to technical labor but must be estimated based on the geographic distribution of clinics across the city's subdistricts and the transport modality used by the technical team.

Technical helpdesk (twelve months). One qualified staff member available throughout follow-up. Cost depends on whether this is a dedicated role or absorbed into an existing research position.

Baseline survey. Administered to enrolled clinics before randomization. Cost depends on the instrument, number of enumerators, per-diem rates, and transport. A formal field budget for this component requires a finalized instrument and clinic list.

Data access. SITB data access is through a formal agreement with Bandung Dinas Kesehatan and the Ministry of Health. Whether this involves financial cost or only administrative time must be clarified before budgeting.

7. Risks and Limitations

7.1 Internal validity

Technical failure. Heryawan and colleagues' catalog of Satu Sehat developer pain points — server issues, data mapping issues, and FHIR profile management issues [16] — makes system

failure a risk. API call logging and a retry protocol are built in to detect such events; technical failures are logged separately and do not count as non-notifications in the primary analysis. A pre-specified sensitivity analysis excludes clinic-months with documented failures.

Avoidance through coding. Providers may shift to vague respiratory ICD codes (J00–J99) to circumvent the automatic trigger for A15–A19. The mechanism check is the explicit test for this. If avoidance is detected, it changes the policy interpretation entirely.

Differential post-assignment behavior. Because the intervention requires an in-person visit, providers cannot be blinded to assignment. Outcome measurement uses administrative SITB records, which removes provider self-report from the outcome but does not eliminate the possibility that treated providers behave differently in ways unrelated to the API. The mechanism check helps adjudicate this: the API mechanism predicts an effect mediated through coded diagnoses; a non-mediated effect would suggest something else is operating.

7.2 External validity

Sample restriction. Eligibility is restricted to EMR-compatible clinics, a more digitally advanced subset of the private sector. Findings will not generalize to paper-based clinics, which remain a separate and arguably larger policy challenge. The 8.9% national EMR adoption figure from the Satrio and Aqid analysis [18] places sharp limits on the immediately addressable population for this intervention.

Geographic specificity. Bandung is a relatively high-infrastructure urban setting. Results may not generalize to rural districts with weaker connectivity or lower Satu Sehat adoption.

Mechanism scope. The intervention addresses administrative friction only. It does not address financial misalignment under JKN capitation [4], patient confidentiality concerns, diagnostic uncertainty, or provider motivation more broadly. These barriers are documented [5,6,7] and may dominate in clinic types or settings outside the trial sample. The mechanism check is intended to indicate which barrier binds in this sample but cannot speak to barriers the design does not vary.

Platform trajectory. Satu Sehat is actively evolving under the Permenkes No. 24/2022 mandate [20]. Findings about technical feasibility and per-clinic implementation cost may not be stable over time as the platform matures.

7.3 Attrition and spillovers

Attrition. Clinics may close, change EMR systems, or withdraw from the trial. The intent-to-treat analysis is primary and includes all randomized clinics regardless of subsequent status. Attrition will be compared across arms. A pre-specified bounding analysis (Lee bounds [23] or equivalent) will be used if differential attrition requires it.

Contamination and spillovers. Control clinics could in principle learn about the intervention from treated clinics in the same neighborhood and adopt elements of it. Subdistrict stratification reduces but does not eliminate this risk. Any contamination would attenuate the estimated effect, biasing β toward zero. Spillover is more concerning if it operates through the Satu Sehat platform itself — for example, if Dinas Kesehatan staff use insights from the trial to push integration to control clinics during the follow-up period. The data-sharing agreement will include a written commitment that no broader integration push occurs in Bandung during the trial follow-up.

8. Conclusion

This Pre-Analysis Plan describes a proof-of-concept cluster-randomized trial of automated, infrastructure-level TB notification in Bandung's private primary care sector. The design's central commitment is to test friction reduction in isolation so that the result has a clean policy interpretation regardless of which way it goes. A positive primary effect with stable TB-specific coding would support scaling the integration approach. A null primary effect or declining TB-specific coding would redirect policy toward behavioral and incentive instruments.

The study is not powered for confirmatory inference. Its purpose is to establish technical feasibility at clinic level, to provide an effect size estimate for a fully powered Stage 2 trial, and to identify which mechanism binds. The key prerequisites before implementation are a precise local EMR adoption figure to finalize the eligible sample, a confirmed staffing model for technical setup, and a formal data-sharing agreement with Bandung Dinas Kesehatan.

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